Machine Learning

PHYS 453

Dr. Daugherity

Abilene Christian University

Part 0

TYPES OF PROBLEMS

Types of ML Problems

1) Supervised Learning
2) Unsupervised Learning
3) Reinforcement Learning

Supervised Learning

Supervised Learning:

- we are given training data where we know the right answers
- pattern recognition finds hidden trends
- training data can be a huge bottleneck

Two main types of supervised learning problems:

Classification: what we've done this semester, place sample in a category

Regression: predict a number. Examples in sklearn are:

- California Housing – predict house prices from size, # of rooms, etc
- Diabetes – measure disease progression based on medical data
- Exercise – predict weight, waist size, pulse rate from exercise data

Unsupervised Learning

We don’t know the right answer. We are given features but no targets.

- **clustering:** can I find natural clusters or groups in the data?
- **generation:** can I make new “realistic” data

Unsupervised: Clustering

sklearn has lots of built in tools for this!

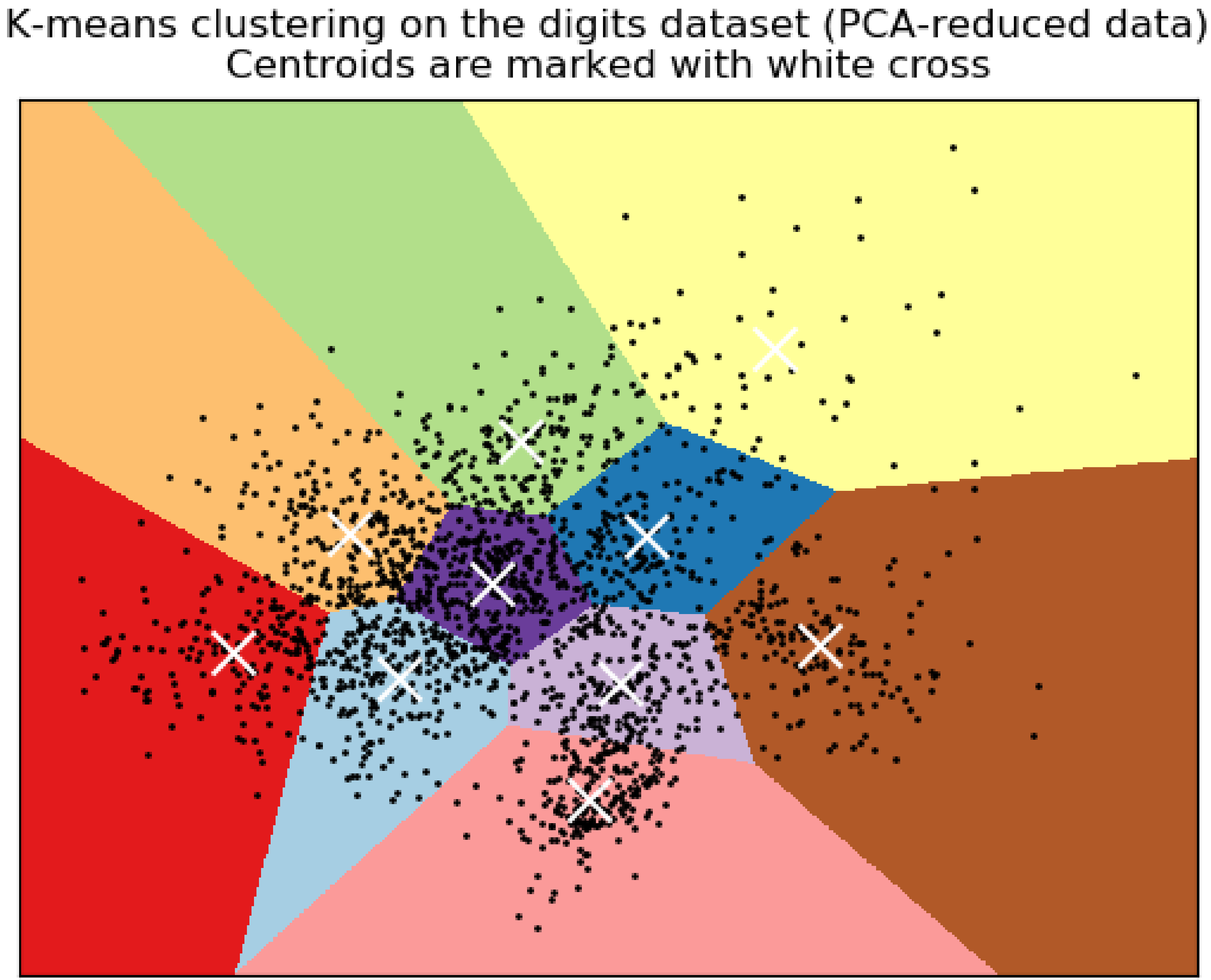
https://scikit-learn.org/stable/unsupervised_learning.html

2. Unsupervised learning

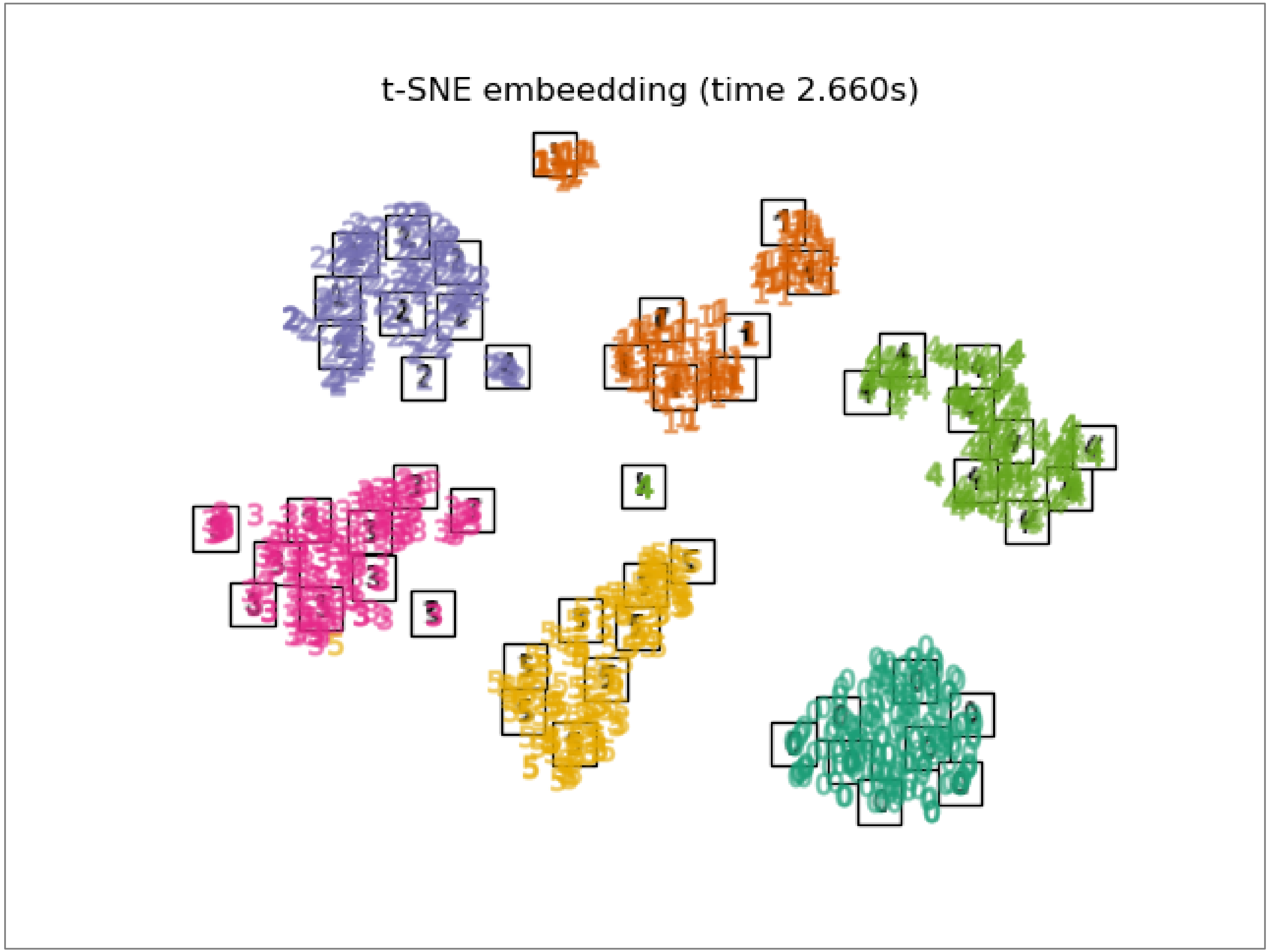
- 2.1. Gaussian mixture models
- 2.2. Manifold learning
- 2.3. Clustering
- 2.4. Biclustering
- 2.5. Decomposing signals in components (matrix factorization problems)
- 2.6. Covariance estimation
- 2.7. Novelty and Outlier Detection
- 2.8. Density Estimation
- 2.9. Neural network models (unsupervised)

k Means Clustering

http://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_digits.html



https://scikit-learn.org/stable/auto_examples/manifold/plot_lle_digits.html



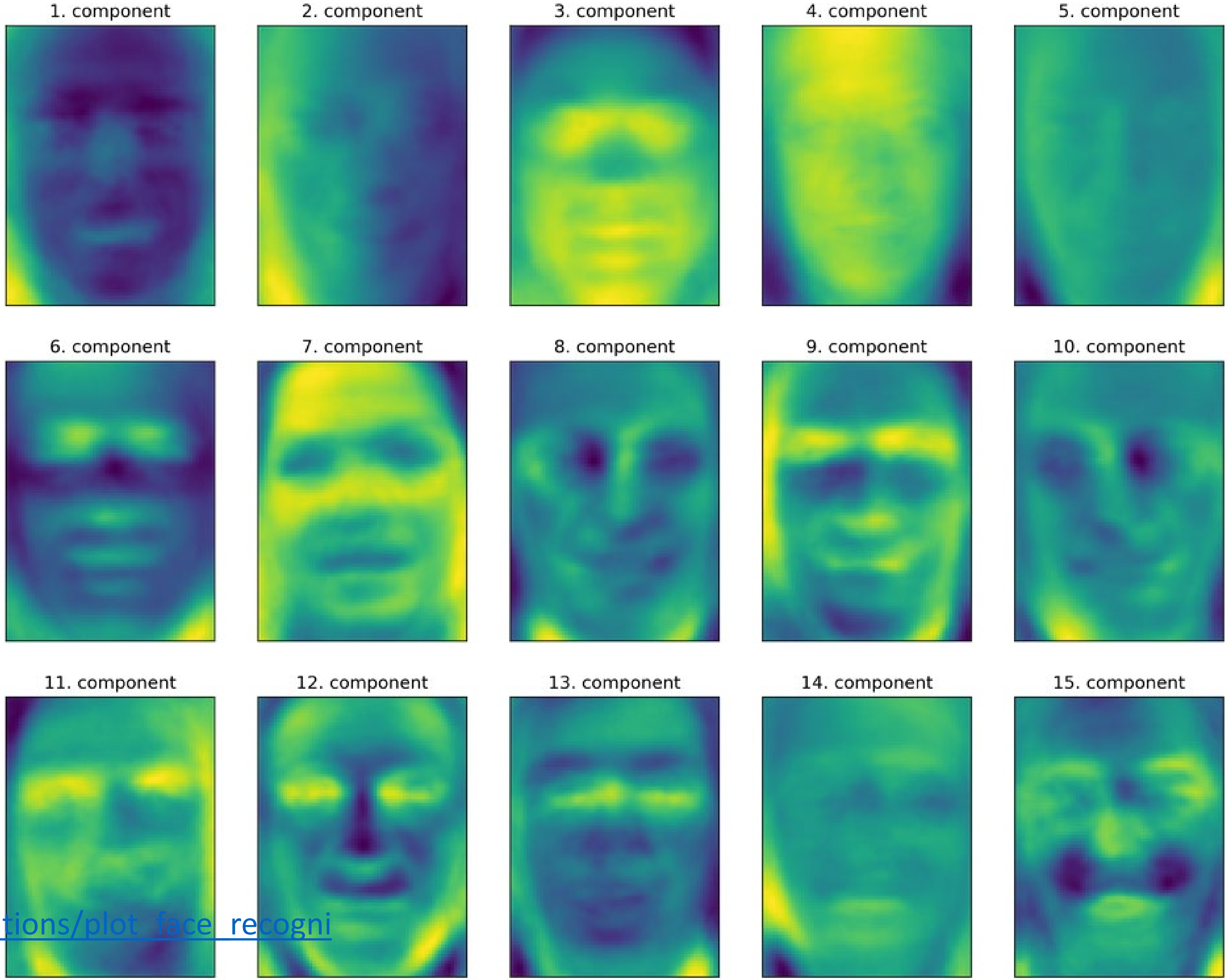
T-SNE on the digits dataset (only using digits 0-5) makes a 2D plot showing clusters of “similar” without knowing the right answers!

Eigenfaces!

If we plot the biggest PCA components for face images we get the eigenfaces!

Every face could be represented as a linear combination of these.

We can also use this trick for completing partial images. Fit the eigenfaces to the portion we've got then predict missing piece



https://scikit-learn.org/stable/auto_examples/applications/plot_face_recognition.html

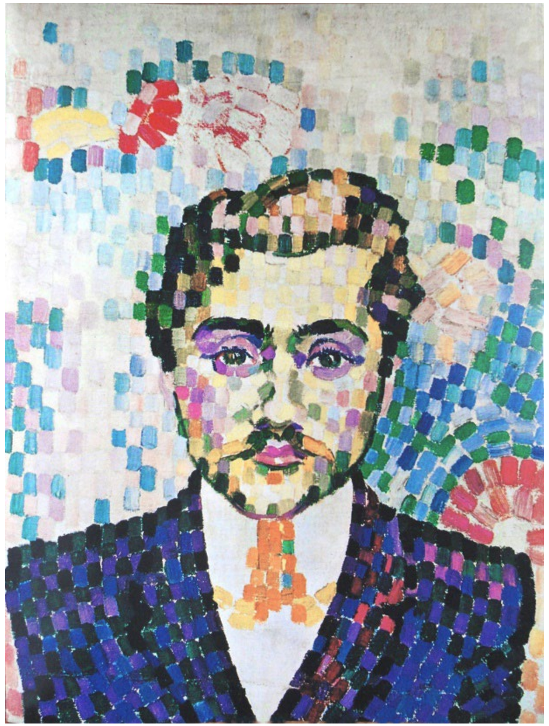
Unsupervised: Generation

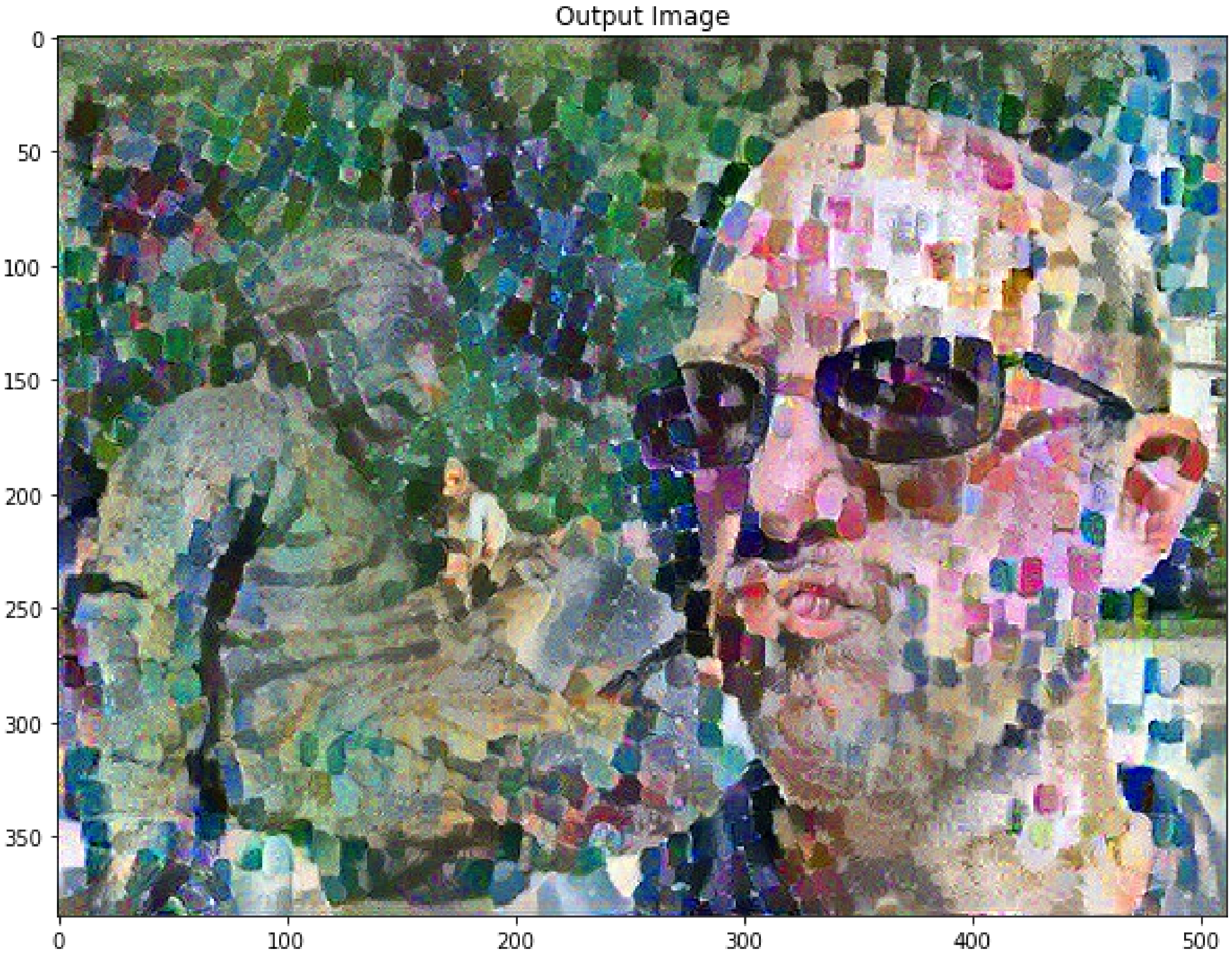
The interesting problems are really, really hard. sklearn doesn't offer much…

Figure 1.5 Generative models for images. Left: two images were generated from a model trained on pictures of cats. These are not real cats, but samples from a probability model. Right: two images generated from a model trained on images of buildings. Adapted from Karras et al. (2020b).

Figure 1.7 Inpainting. In the original image (left), the boy is obscured by metal cables. These undesirable regions (center) are removed and the generative model synthesizes a new image (right) under the constraint that the remaining pixels must stay the same. Adapted from Saharia et al. (2022a).

Figure 1.12 Multiple images generated from the caption "A teddy bear on a skateboard in Times Square." Generated by DALL·E-2 (Ramesh et al., 2022).




Output Image
0
50
100
150
200
250
300
350
0
100
200
300
400
500

Reinforcement

- we have an agent that can take actions
- establish reward
- learn how to choose actions that lead to high rewards

Examples:

- learn to make a robot walk
- social media algorithm
- learn to maximize score on a game

https://docs.pytorch.org/tutorials/intermediate/reinforcement_q_learning.html

Reinforcement Learning (DQN) Tutorial

Created On: Mar 24, 2017 | Last Updated: Jun 16, 2025 | Last Verified: Nov 05, 2024

Author: Adam Paszke

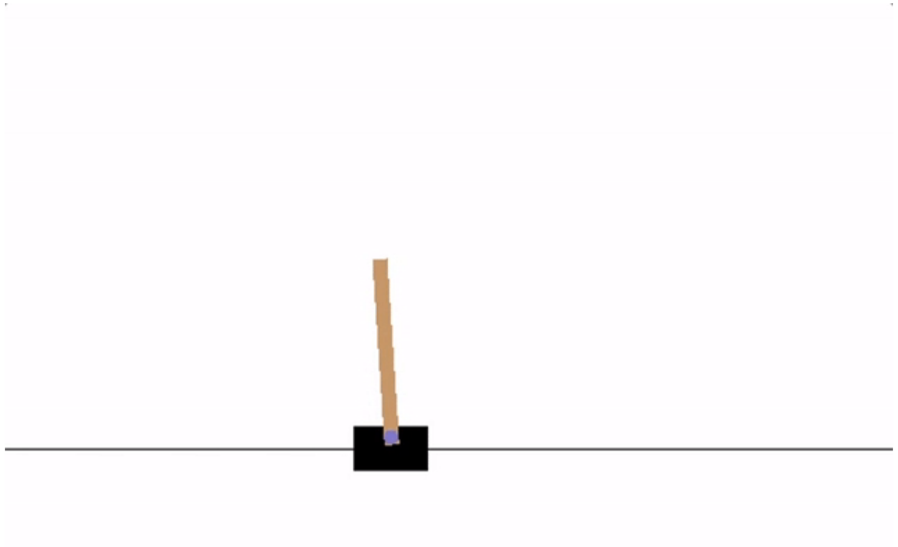
Mark Towers

This tutorial shows how to use PyTorch to train a Deep Q Learning (DQN) agent on the CartPole-v1 task from Gymnasium.

You might find it helpful to read the original Deep Q Learning (DQN) paper

Task

The agent has to decide between two actions - moving the cart left or right - so that the pole attached to it stays upright. You can find more information about the environment and other more challenging environments at Gymnasium's website.



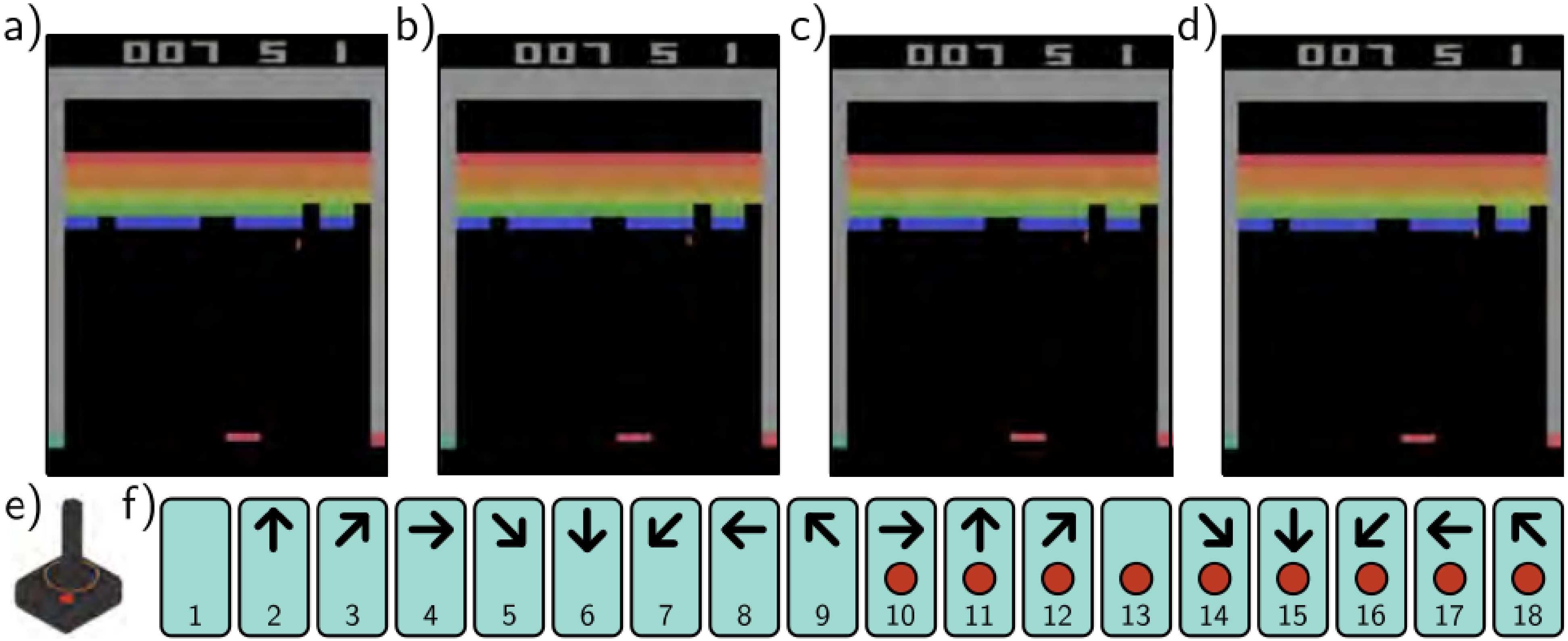


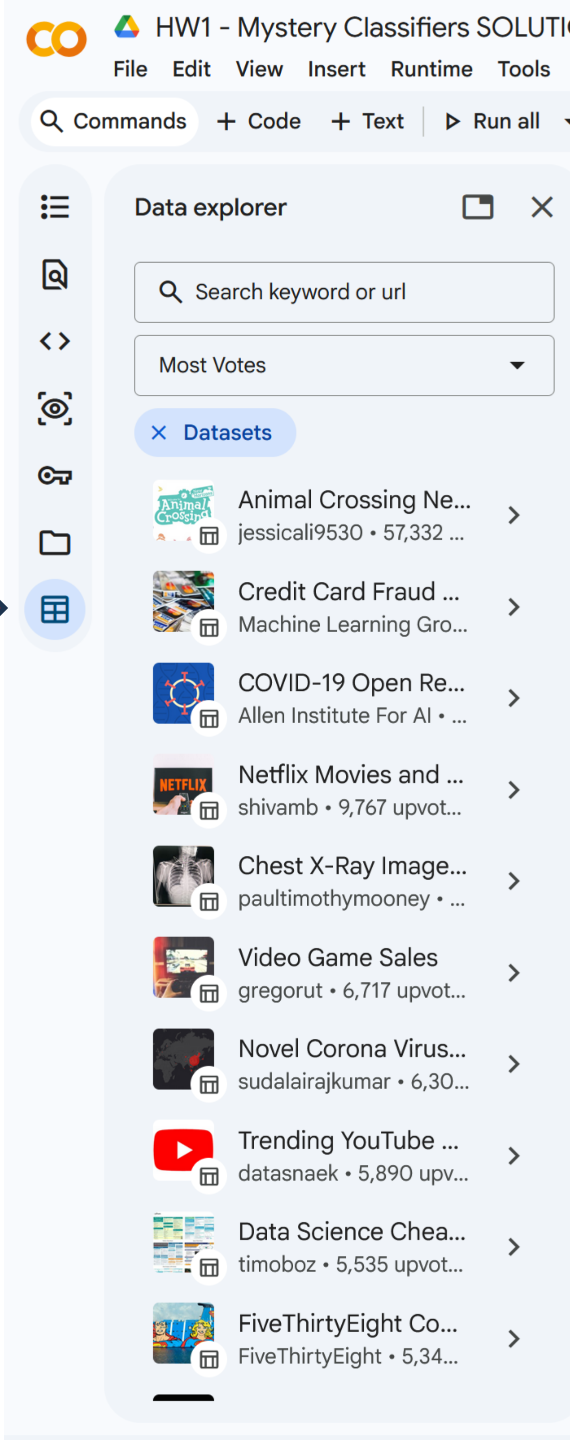
Figure 19.13 Atari Benchmark. The Atari benchmark consists of 49 Atari 2600 games, including Breakout (pictured), Pong, and various shoot-em-up, platform, and other types of games. a-d) Even for games with a single screen, the state is not fully observable from a single frame because the velocity of the objects is unknown. Consequently, it is usual to use several adjacent frames (here, four) to represent the state. e) The action simulates the user input via a joystick. f) There are eighteen actions corresponding to eight directions of movement or no movement, and for each of these nine cases, the button being pressed or not.

Part 1

SOURCES OF DATA

Links to Data

- https://www.openml.org/search?type=data&status=active
- https://www.kaggle.com/datasets
- https://huggingface.co/datasets
- Google Colab Data explorer



Types of Data

- tabular
- text
- images
- audio

DIY

never forget that you can make your own, but it will be harder than you think…

It is cool to have something in the real world

Part 2

TOOLS

pytorch

The most popular and powerful tool by far for advanced deep learning is pytorch

- wish I had a second semester to do this…
- tedious and painful to get started, lots and lots of lines of "boilerplate" needed to do anything
- some tools exist to make it easier, but you may be better off just doing straight pytorch
  - skorch: brings pytorch into sklearn https://github.com/skorch-dev/skorch
  - lightning
  - ignite

Part -1

THINGS TO CONSIDER

To Consider

- month-long project
- source of data
- using unfamiliar tools
- awesomeness
- portfolio